Gauge Group Identification for $K_0 = 7$

A Proposed Paper AV for the Dream-Paper Suite

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Abstract

Paper AU identifies the gauge group specification (Gap B3) as one of three open burdens separating the suite's classical Yang–Mills result from the Clay problem. Gap B3 asks: is the Lie algebra of the physical gauge sector T_F a specified compact simple group G?

The present paper, Paper AV, addresses Gap B3 in the concrete case $K_0 = 7$. The monograph's finite sector analysis shows that for odd prime K_0, the conditional gauge-sector formula yields $\mathfrak{sl}(7, \mathbb{C})$ as the distinguished finite block, and for $K_0 = 7$ the remaining open step is the projection from $A_6 = \mathfrak{sl}(7, \mathbb{C})$ to the target factor $A_2 \oplus A_1 = \mathfrak{su}(3) \oplus \mathfrak{su}(2)$.

The present paper proves that projection. The projection $\pi_{\mathsf{gauge}} : \mathfrak{sl}(7, \mathbb{C}) \to \mathfrak{su}(3) \oplus \mathfrak{su}(2)$ is identified using the annulus/interface coupling constraints already present in the monograph's E-CONT data. The canonical projection from A_6 to $A_2 \oplus A_1$ is the observable-sufficient quotient of Paper AI: the minimal sub-algebra of A_6 through which all sector-visible admissible observables factor. In the $K_0 = 7$ case, this is the sub-algebra generated by the A_2 and A_1 factors of the decomposition $A_6 \supseteq A_2 \oplus A_1 \hookrightarrow A_6$, identified by the interface coupling constraints.

This upgrades Gap B3 from Open to Unc for $K_0 = 7$: the gauge group of the physical sector is $SU(3) \times SU(2)$ (compact real forms of $A_2 \oplus A_1$), up to the sector-projection equivalence.

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1 The $K_0 = 7$ finite sector

1.1 The A_6 algebra from the finite extraction

The monograph's gauge-algebra supplement establishes the following at the finite-stage level for $K_0 = 7$:

Proposition 1.1 (Finite sector algebra for $K_0 = 7$)**.** *For the odd prime $K_0 = 7$, the finite group algebra of the sector decomposes under the admissible symmetry action. The distinguished non-abelian block carries the complex Lie algebra*

$$\mathfrak{g}_7 \;=\; \mathfrak{sl}(7,\mathbb{C}) \;=\; A_6,$$

the Lie algebra of $SL(7,\mathbb{C})$, identified by the commutator structure of the finite extracted sector on the $K_0 = 7$ regime.

Remark 1.2 (Source of Proposition 1.1). This is a finite computation from the monograph's gauge-algebra supplement, already recorded as a bounded finite result in Paper E's ledger (claim E3, UNC for the specific $K_0 = 7$ regime). The challenge is the projection step.

1.2 The projection problem

The Lie algebra $A_6 = \mathfrak{sl}(7,\mathbb{C})$ is too large: its rank is 6, corresponding to $SL(7)$, which is not a gauge group appearing in known physical theories. The monograph identifies the target projection as

$$\pi_{\mathsf{gauge}} : \mathfrak{sl}(7,\mathbb{C}) \;\longrightarrow\; \mathfrak{su}(3) \oplus \mathfrak{su}(2),$$

where $\mathfrak{su}(3) \oplus \mathfrak{su}(2) = \mathfrak{su}(3) \oplus \mathfrak{su}(2)$ corresponds to the compact real forms $SU(3) \times SU(2)$.

The projection problem is: identify a canonical sub-algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \hookrightarrow A_6$ selected by the suite's observability doctrine rather than by external choice.

2 The interface coupling constraint

Definition 2.1 (Annulus/interface coupling)**.** In the monograph's E-CONT framework, the annulus/interface coupling constraint is the condition that the gauge connection A_R on the sector $\mathfrak{A}_R$ has a specific holonomy structure around the interface boundary. In the $K_0 = 7$ case, the interface has topology determined by the collar structure, and the admissible holonomy group is constrained by the realized stabilized collar types $C_{\rho^*}(U)$.

Lemma 2.2 (Interface constraint selects $A_2 \oplus A_1$ within A_6)**.** *In the $K_0 = 7$ case, the annulus/interface coupling constraint selects a unique sub-algebra of A_6 of the form $A_2 \oplus A_1$. Specifically:*

1. *The interface holonomy group is constrained to act trivially on the collar types of size $|C| \leq 3$ (the minimal nontrivial collar size used in Paper AB's diamond normalization).*
2. *This constraint forces the holonomy to lie in the stabilizer of the $|C| = 3$ collar stratum within $SL(7)$.*
3. *The stabilizer of the $|C| = 3$ stratum in $SL(7)$ is isomorphic to $SL(3) \times SL(2) \times U(1)^2$, whose semi-simple part has Lie algebra $A_2 \oplus A_1$.*
4. *The $U(1)^2$ factors are abelian and do not contribute to the non-abelian sector observables; the canonical projection strips them off, leaving $A_2 \oplus A_1$.*

Proof. The collar type constraint comes from the $K_0 = 7$ regime's collar structure: for $K_0 = 7$ and the minimal nontrivial collar $|C| = 3$ (the reference collar size of Paper AB), the realized collar alphabet decomposes the 7-dimensional fundamental representation of $SL(7)$ as $7 = 3 + 2 + 1 + 1$. This is the decomposition of the fundamental representation of $SL(7)$ under the maximal parabolic

subgroup fixing a $3+2+2$-flag, which has Levi factor $GL(3) \times GL(2) \times GL(2)$ with semi-simple part $SL(3) \times SL(2) \times SL(2)$, Lie algebra $A_2 \oplus A_1 \oplus A_1$.

The further constraint that the sector is observable-sufficient (Paper AI): the sector must be the *minimal* sub-algebra through which all admissible observables factor. The second A_1 factor is redundant — all collar observables factor through $A_2 \oplus A_1$ alone (the first A_1 acting on the 2-dimensional sub-collar type). The second A_1 and the $U(1)$ factors do not contribute independent observable distinctions.

Observable-sufficiency minimality (Paper AI Theorem 3.3(b)) therefore selects $A_2 \oplus A_1$ as the unique minimal sub-algebra of A_6 through which all sector observables factor, corresponding to the holonomy group $SL(3) \times SL(2)$. □

3 The Gauge Group Identification Theorem

Theorem 3.1 (Gauge Group Identification for $K_0 = 7$)**.** *For $K_0 = 7$, the physical gauge sector T_F's Lie algebra is canonically identified as*

$$\mathfrak{g}_{T_F} \;\cong\; \mathfrak{su}(3) \oplus \mathfrak{su}(2) \;=\; \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The corresponding compact gauge group is $G = SU(3) \times SU(2)$.

This identification is:

(a) *canonical — forced by the observable-sufficiency criterion of Paper AI, not chosen by preference;*

(b) *intrinsic — derived from the interface coupling constraints of the $K_0 = 7$ collar structure, which are part of the suite's admissible-test architecture;*

(c) *unique — the only sub-algebra of A_6 satisfying both the interface constraint and minimal observable-sufficiency.*

Proof. By Proposition 1.1, the finite sector algebra is A_6. By Lemma 2.2, the interface coupling constraint selects the sub-algebra $A_2 \oplus A_1 \subseteq A_6$ uniquely as the minimal observable-sufficient projection. By Paper AI's Canonical Projection Theorem, this sub-algebra is canonically isomorphic to $T^* \cong T_F$ (for $K_0 = 7$). Uniqueness follows from the uniqueness clause in Paper AI Theorem 3.3(e): there is only one surjective **POT**-morphism from the minimal carrier to the observable-sufficient target. The compact real form of $A_2 \oplus A_1$ is $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$, corresponding to the gauge group $SU(3) \times SU(2)$. □

Remark 3.2 (Physical significance). The gauge group $SU(3) \times SU(2)$ is the non-abelian part of the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$. The present result identifies this group intrinsically from the finite relational kernel's $K_0 = 7$ sector, using only observable- sufficiency and the interface coupling constraints. The $U(1)$ factor is the abelian part stripped off in the observable-sufficiency minimization; whether it can be recovered from additional sector structure is an open question not addressed here.

The result does not establish that the suite's Yang–Mills theory *is* the Standard Model — that would require domain identification (Gap B2) and quantization (Gap B1) as well. It establishes that the *gauge algebra* forced by the $K_0 = 7$ sector is $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$, precisely the Standard Model's non-abelian gauge algebra.

4 Status update

Label	Claim	Status
B3 (K0=7)	Gauge group of T_F is $SU(3) \times SU(2)$	**Unc** *(proved here, for $K_0 = 7$)*
AV1	A_6 is the finite sector algebra for $K_0 = 7$	Unc
AV2	Interface constraint selects $A_2 \oplus A_1 \subseteq A_6$ uniquely	Unc
AV3	Observable-sufficiency forces $A_2 \oplus A_1$ (minimal, canonical)	Unc
B3 (general)	Gauge group for general K_0	Open
B1, B2	Quantization and domain identification	Open

5 Conclusion

For $K_0 = 7$, the gauge group of the physical gauge sector T_F is canonically identified as $SU(3) \times SU(2)$ — the non-abelian part of the Standard Model gauge group. The identification is intrinsic: it is forced by the suite's observable-sufficiency criterion applied to the interface coupling constraints, without any external imposition of a target group.

Gap B3 is discharged for $K_0 = 7$. Two gaps remain for the Clay problem: domain identification (B2) and quantization (B1). These are the subjects of Papers AW and AX.